

TRUE PHYSICS

(unknown to the solver)

$$\frac{\partial c}{\partial t} = \nabla \cdot (D \nabla c) - R(c)$$

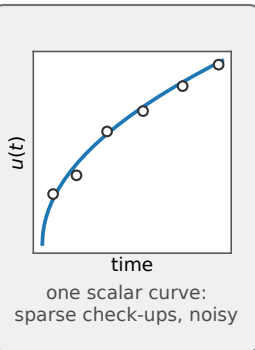
moving boundary $\delta(t)$

rate constants $D, k, \dots$

solid-electrolyte interphase (SEI)
or passive-film growth

observe

WHAT YOU MEASURE



fit
 $\mathcal{D}^{(M)} u = f$

WHAT SURVIVES



effective order α
robust to 5% noise



tempered (α, λ)
needs several conditions



spectrum $w(\alpha)$
not identifiable

NOT recovered: the governing PDE • the kernel shape • the microscopic rate constants

Inversion collapses to a low-dimensional memory descriptor (at most an effective order).